

# Project Executable Files & Execution Guide

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|--------------|------------------------------------------------------|
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| Project Name | CaptionAI: AI-Powered Social Media Caption Generator |

## Project Executable Files & Execution Guid

To evaluate this project, evaluator require access of source code and live deployments etc. which were provided below:

### Project Links & Access

| Resource Type                     | URL / Access Link                                                                                                                                                                          |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source Code Repository (GitHub)   | <a href="https://github.com/khushboomaher23-beep/Caption-AI"><u>https://github.com/khushboomaher23-beep/Caption-AI</u></a>                                                                 |
| Live Deployed Application (Ngrok) | <a href="https://tabasco-renovator-sandpit.ngrok-free.dev"><u>https://tabasco-renovator-sandpit.ngrok-free.dev</u></a>                                                                     |
| Demo Video Link                   | <a href="https://drive.google.com/file/d/1XBF0lHL5a9QyvbFi1SlrDOrqzrPb2exZ/view?usp=sharing"><u>https://drive.google.com/file/d/1XBF0lHL5a9QyvbFi1SlrDOrqzrPb2exZ/view?usp=sharing</u></a> |

## Key Executable Files & Folder Structure

The primary directories and the core execution files were located within my repository.

| S. No. | Folder / File Name | Description & Purpose                                                                                                                                                                |
|--------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | app.py             | The main Python file that starts the Flask application. It manages routing, processes user requests, and connects the application with the Groq API for AI-based caption generation. |

|   |                             |                                                                                                                                                                                 |
|---|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>templates/index.html</b> | The main user interface of the application. It contains the input form, platform selection, tone selection, and the section where generated captions are displayed.             |
| 3 | <b>static/js/main.js</b>    | Handles client-side functionality such as form submission, loading animation, API requests, and the copy-to-clipboard feature.                                                  |
| 4 | <b>static/css/style.css</b> | Defines the appearance of the application, including layout, colors, fonts, buttons, and responsive design for different screen sizes.                                          |
| 5 | <b>.env</b>                 | Stores environment variables such as the Groq API key. Keeping sensitive information in this file helps improve security and prevents it from being exposed in the source code. |
| 6 | <b>requirements.txt</b>     | Lists all Python libraries required to run the project, including Flask, Groq, and python-dotenv.                                                                               |

### Step 3: Local Setup & Execution Instructions

Step-by-step guide to installing dependencies and launching the application on a local machine.

| Step | Action / Command                                                                                                                                   | Expected Outcome                                                                       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1    | <code>git clone <a href="https://github.com/khushboomaher23-beep/Caption-AI.git">https://github.com/khushboomaher23-beep/Caption-AI.git</a></code> | Downloads the complete project source code to the local computer.                      |
| 2    | <code>python -m venv venv</code><br><code>venv\Scripts\activate</code><br>(Windows)                                                                | Creates and activates a virtual environment to manage project dependencies separately. |
| 3    | <code>pip install -r requirements.txt</code>                                                                                                       | Installs all required Python packages listed in the requirements file.                 |

|   |                                      |                                                                                                                                              |
|---|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | Configure the <code>.env</code> file | Add the required Groq API key and other environment variables before running the application.                                                |
| 5 | <code>python app.py</code>           | Starts the Flask application locally. By default, the application runs at <u><a href="http://127.0.0.1:5000/">http://127.0.0.1:5000/</a></u> |
| 6 | <code>ngrok http 5000</code>         | Creates a public URL that allows others to access the locally running application for testing or demonstration purposes.                     |